CxRisk: Cervical Cancer Risk Stratification and Screening Triage

94-706 Healthcare Information Systems – Spring 2026
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Agenda

1	Context & Motivation
2	Data: Sources, Analysis, DB
3	DSS Architecture
4	Recommendations & Conclusion
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Context & Motivation

Problem Statement

Cervical cancer screening is preventable and guideline-based, but triage still depends heavily on manual chart review, clinician judgment, and finite testing/referral capacity.

Clinical Context & Motivation

Why this matters

Cervical cancer is **largely preventable** through HPV vaccination and screening, but screening and follow-up remain uneven across populations.

Operational gap

Clinics must triage patients with **incomplete histories**, finite HPV/Pap capacity, and competing referral demand.

DSS opportunity

A DSS can summarize risk factors, **standardize triage logic**, and surface patients needing faster follow-up.

- CDC screening guidance emphasizes routine cervical cancer screening for eligible patients; national data show **screening remains below universal coverage**.
- The prototype targets the step before or during screening/referral decisions, where **chart review is often fragmented and time constrained**.
- The DSS is **advisory**: it estimates risk, shows drivers and missing-data caveats, recommends a next action, and preserves clinician override.

Gaps in Current Solutions

ASCCP Risk-Based Management Consensus Guidelines

- Determines a patient's immediate risk of precancer (CIN3+) or 5-year risk of progressing to cancer (BD, 2022)
- Limited by requiring a prior lab test (HPV/Pap) to conduct analysis, leaving out patients who may be at risk due to behavioral factors
- **Is not integrated into a clinician's EHR system**

Improving lives through the prevention and treatment of anogenital & HPV-related diseases

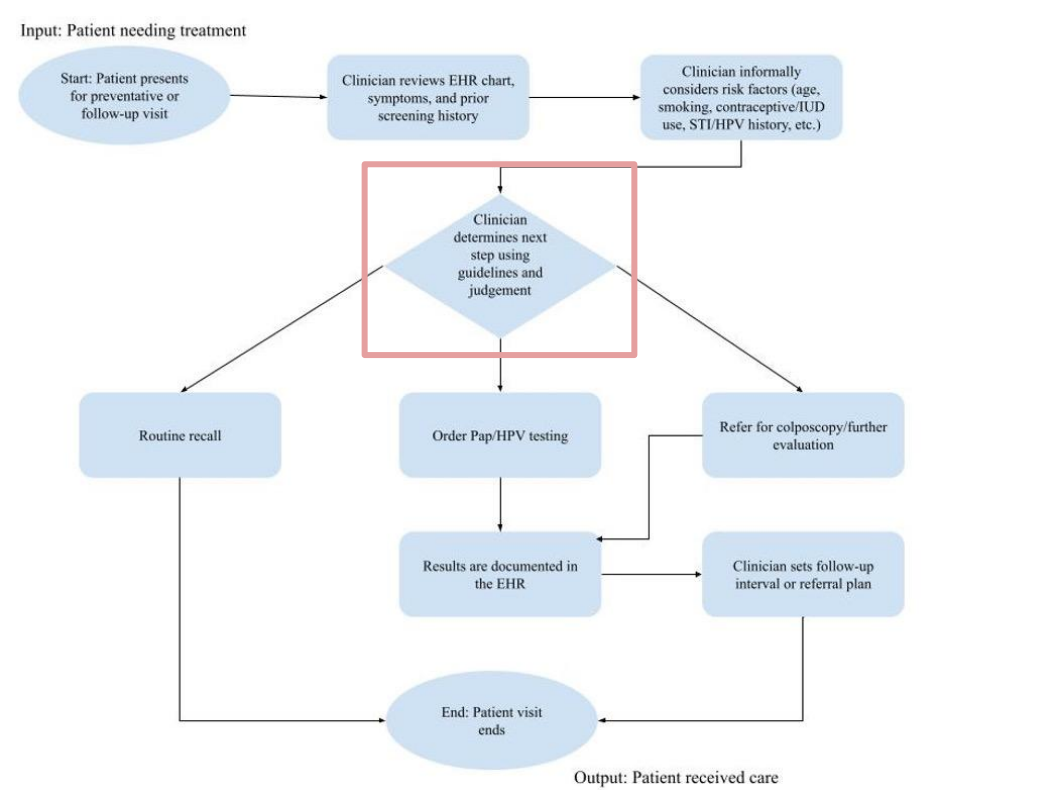
Benefits of an IT Solution in a Clinical Workflow

- Risk aggregation
  - Structured risk factors are entered/reviewed once and converted into a consistent patient-level score.
- Explainability in recommendations
  - The UI shows the top drivers and missing-data flags rather than a black-box recommendation.
- Workflow coordination
  - Nurses and care coordinators can track testing, referral, outreach, and unresolved follow-up.
- Population management
  - Managers can monitor high-risk cohorts, referral demand, overrides, and common risk drivers.
- Document support
  - The tool can generate auditable triage outputs and support chart-note or patient-message drafts.

Project Goal

1. Predicted probability of an abnormal screening/diagnostic outcome
2. Recommended next action: routine recall, expedited HPV/Pap testing, or referral
3. Top risk drivers and clinician override support to preserve judgment

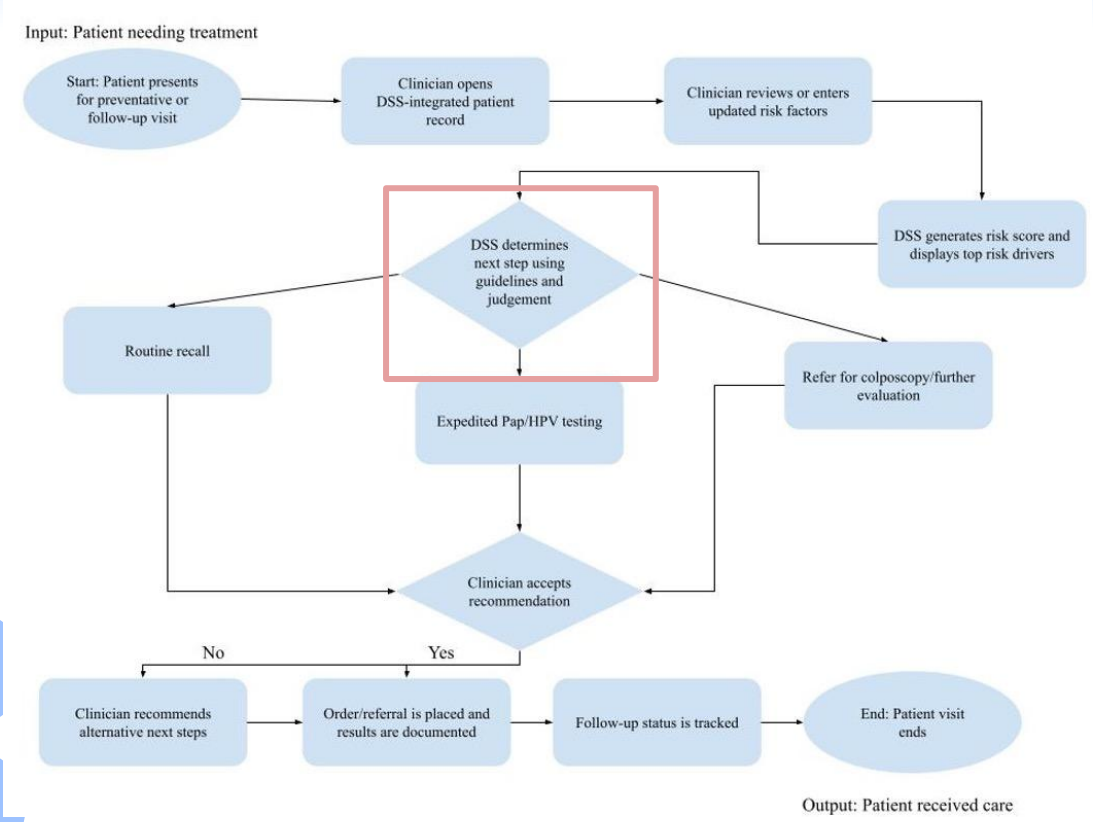
Status Quo Workflow



Current Pain Points:

- No standardized risk score before screening or referral decisions
- Manual chart review consumes visit time
- Variation across clinicians can create inconsistent follow-up
- Capacity constraints are not explicitly visible in the triage step

DSS-Enabled Workflow



What changes?

- Structured risk factors are entered or reviewed in one place
- The DSS computes a patient-level risk estimate and shows top drivers
- Recommendation becomes explicit, auditable, and easier to monitor
- Clinician can accept or override the recommendation with reasoning
- Prototype triage policy
  - Low predicted risk → routine recall
  - Intermediate risk → expedited HPV/Pap
  - Higher risk / concerning pattern → referral

Identified Stakeholders

- Clinicians
  - Physicians (OB-GYNs, PCPs)
    - Need a quick risk estimate at the point of care
    - Want transparent drivers, not a black box
    - Need override capability and documentation support
  - Nurses
    - Need follow-up status, scheduling, and outreach visibility
    - Need to see who needs testing or referral next
- Administrators
  - Staff
  - Managers
  - Quality Team
- Need population-level view of demand and common risk factors
- Need referral volume and override monitoring

Data: Sources, Analysis, DB

Data Source

Cervical Cancer Dataset sourced from the UCI Machine Learning Repository

- 858 patient instances each with 32 features
- 4 binary classification labels to indicate presence of cancer depending on different tests conducted. The behavioral risks in the dataset include smoking, pregnancies, number of sexual partners, use of hormone contraceptives, etc., along with clinical information regarding specific strains of STDs to obtain a clear picture of the patient's history.

Dataset and Preprocessing Pipeline

Source data

- Uploaded cervical cancer risk-factor CSV used as the project prototype dataset
- 835 records × 36 original variables
- Predictors span age, sexual history, smoking, contraceptive/IUD use, STI indicators, and prior diagnosis flags
- Outcomes include Hinselmann, Schiller, Citology, and Biopsy

Cleaning outputs

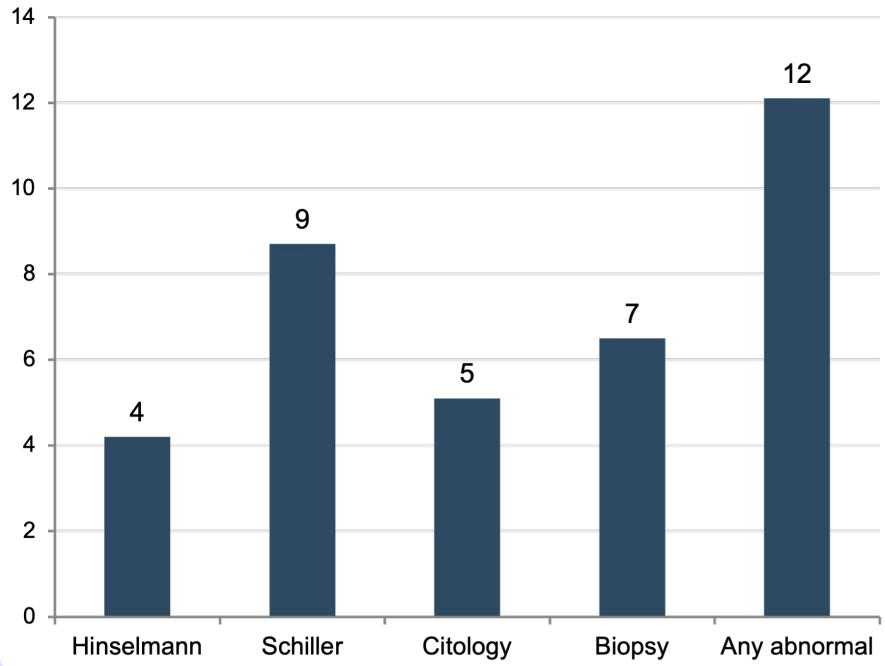
- clean_nullable: 835 × 38
- analysis_ready_imputed: 835 × 67

1. Normalize column names to snake_case; remove exact duplicates
2. Add derived fields: abnormal_any and missing_count_predictors
3. Preserve NULLs in the cleaned dataset for database staging
4. For analysis-ready data: create missingness indicators, mode-impute binary variables, median-impute continuous variables
5. Drop only the two predictors with >90% missingness from the imputed modeling file

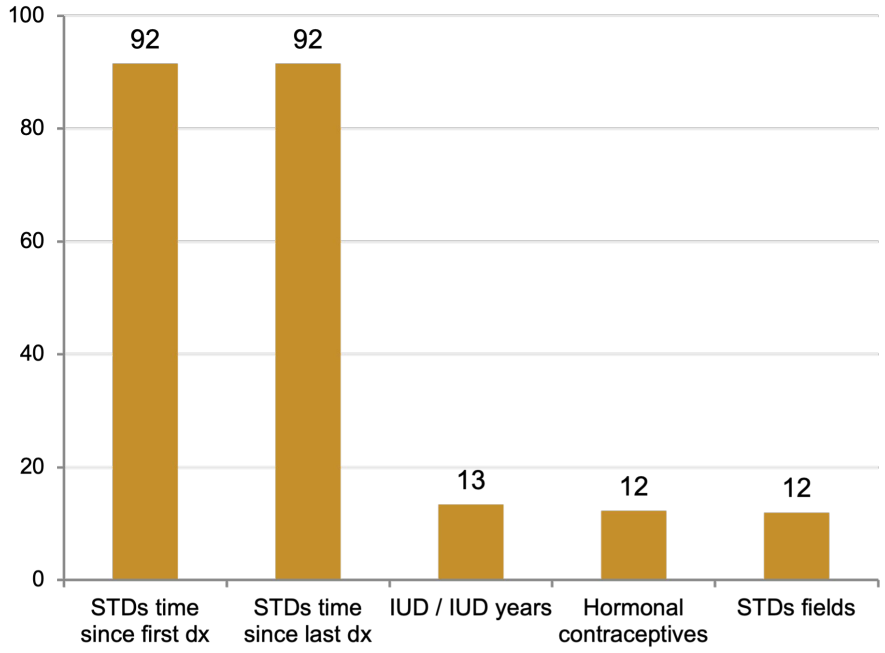
Dropped from imputed file only: stds_time_since_first_diagnosis and stds_time_since_last_diagnosis

Exploratory patterns from the dataset

Outcome Prevalence



Missingness Profile



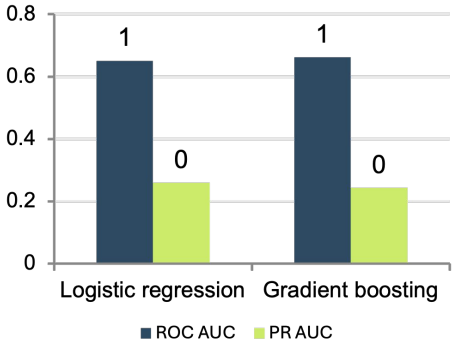
***Key takeaways**: the dataset is class-imbalanced (12.1% any abnormal) and strongly affected by missingness in a small number of STI timing fields, which supports the project emphasis on explicit missing-data handling and sensitivity analysis.*

Modeling Approach and Early Findings

Prototype models

- Compare logistic regression with gradient boosting, consistent with the proposal
- Use analysis-ready imputed dataset with explicit missingness indicators
- Primary presentation target: abnormal_any (positive if any screening/diagnostic outcome is abnormal)
- Operationally, threshold selection should balance missed cases against over-referral under clinic capacity limits

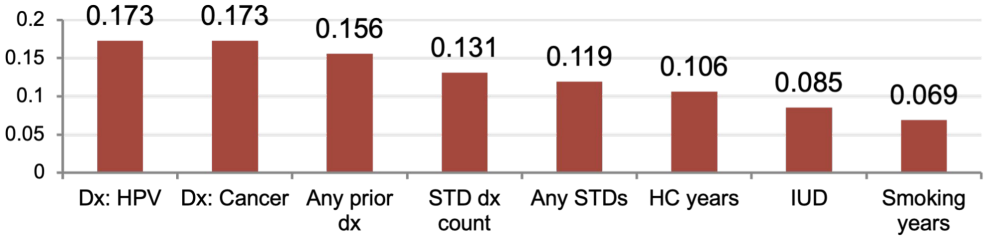
5-fold CV performance



Interpretation

Gradient boosting slightly **improves ROC AUC**, while logistic regression is easier to explain and achieved stronger recall at a balanced threshold. That makes LR a strong baseline for a clinician-facing prototype.

Most associated features with abnormal_any



The Model

Model Performance And Trustworthiness

0.768	18%	0.584	0.581
Holdout AUC	Recall Gain	OOF Ensemble AUC	Confidence Gap

Weighted Ensemble → Final Prediction + Confidence Score

Training: 5-fold stratified CV · Isotonic calibration · SMOTE for class imbalance · Feature: Pregnancies, Sexual Partners, Hormonal Contraceptives, Age, STD burden, IUD use, Smoking

835 Training Patients	172 Holdout Patients	6.4% Positive Rate	5-Fold Cross Validation

Architecture

Modeling Approach and Architecture

Four base learners combined in a weighted ensemble — designed for clinical interpretability

LR

Logistic Regression

Linear baseline. Interpretable coefficients, stable under class imbalance. Weight: 0.15

OOF AUC: **0.543**

RF

Random Forest

Bagged trees. Robust to outliers, captures non-linear interactions. Weight: 0.75

OOF AUC: **0.593**

XGB

XGBoost

Gradient boosted trees. Handles complex feature interactions. Weight: 0.01

OOF AUC: **0.523**

LGB

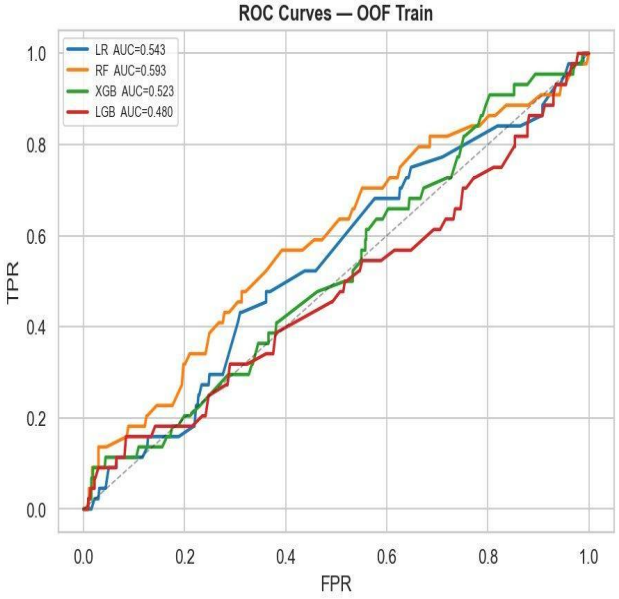
LightGBM

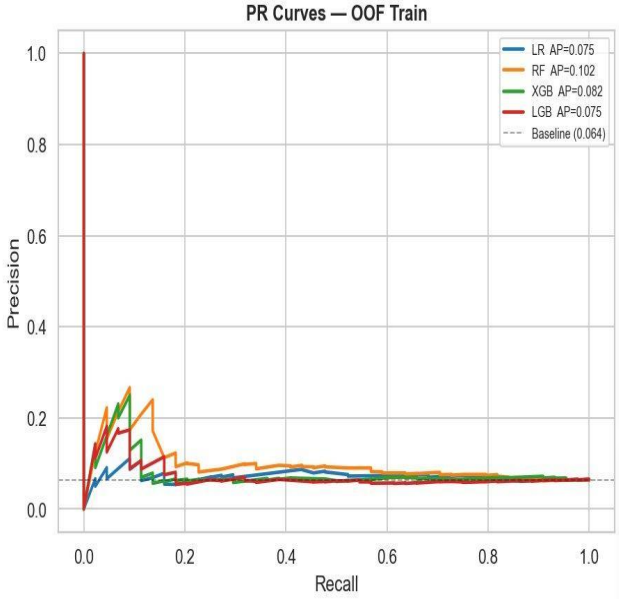
Fast gradient boosting with leaf-wise growth. Low memory footprint. Weight: 0.09

OOF AUC: **0.480**

Architecture

Out-of-Fold Performance: ROC & Precision-Recall





RF leads OOF

Random Forest achieves best OOF AUC (0.593) and best Average Precision (0.102).

Ensemble dominates

Ensemble AUC 0.584 beats LR, XGB, LGB — aggregation reduces variance.

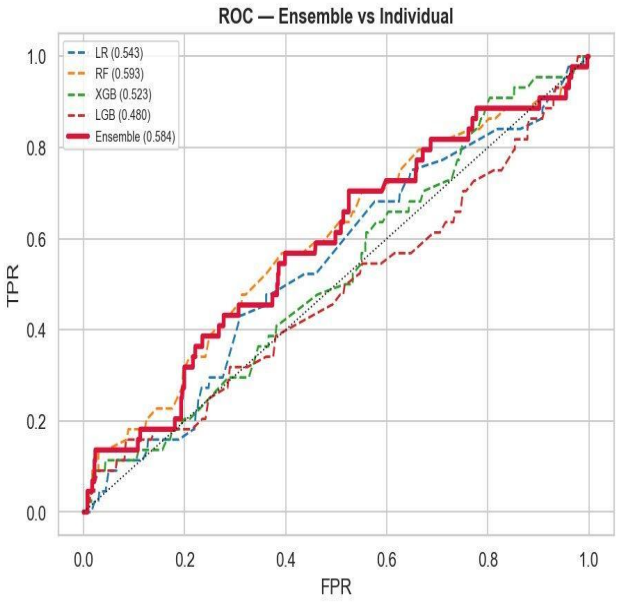
Imbalanced task

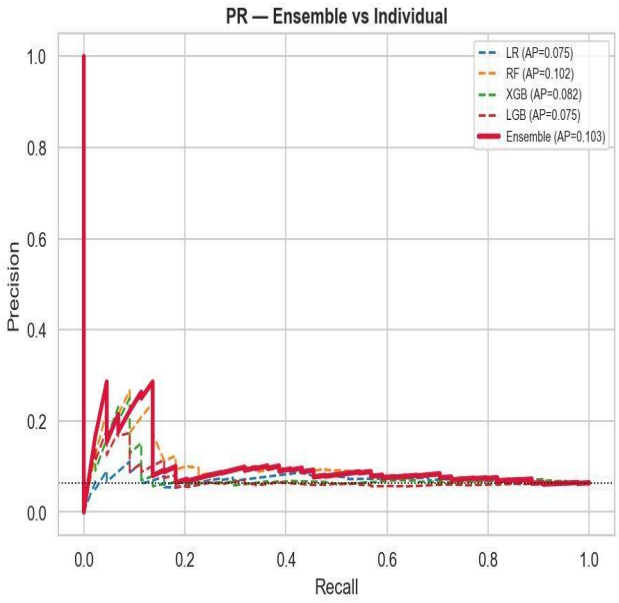
PR-AUC baseline is 0.064 (prevalence). All models exceed this benchmark, with ensemble at 0.103.

Evaluated on held-out folds during training — honest estimate of generalization

OOF Performance

Ensemble vs. Individual Models





The ensemble consistently outperforms any single model — diversity drives gains

Why Ensemble Wins

1. Error diversity

Each model fails on different patients — combining predictions reduces correlated errors.

2. AUC: 0.584

Ensemble AUC beats LGB (0.480), XGB (0.523), LR (0.543). Only RF (0.593) is close.

3. AP: 0.103

Best precision-recall area under curve — critical for this imbalanced clinical task.

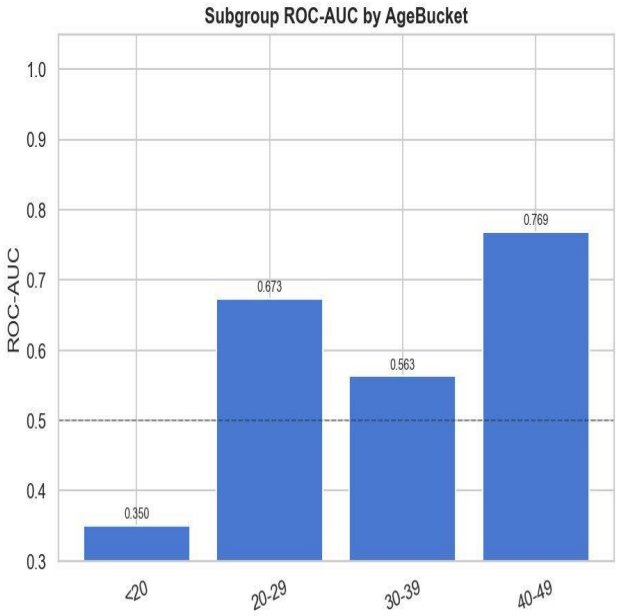
4. Weighted blend

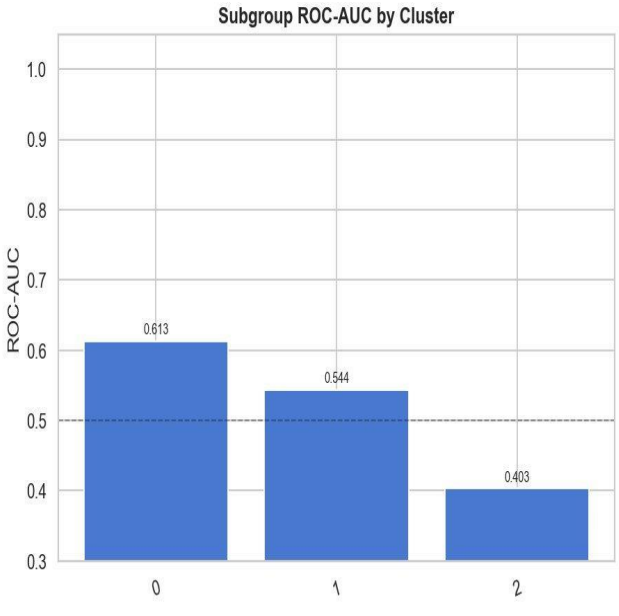
RF gets 75% weight (best performer). LR 15%, LGB 9%, XGB 1% — data-driven weights.

Ensemble Value

Fairness & Subgroup Performance

AUC by age bucket and patient cluster — ensuring equitable triage across populations





Equity monitoring is a live DSS feature — override rates and AUC are tracked by subgroup in the population dashboard.

Fairness Findings

<20 (AUC 0.35)

Below chance — teen patients need tailored features or separate model.

20–29 (AUC 0.67)

Solid performance in the largest screening cohort.

40–49 (AUC 0.77)

Best subgroup — older patients have richer clinical histories.

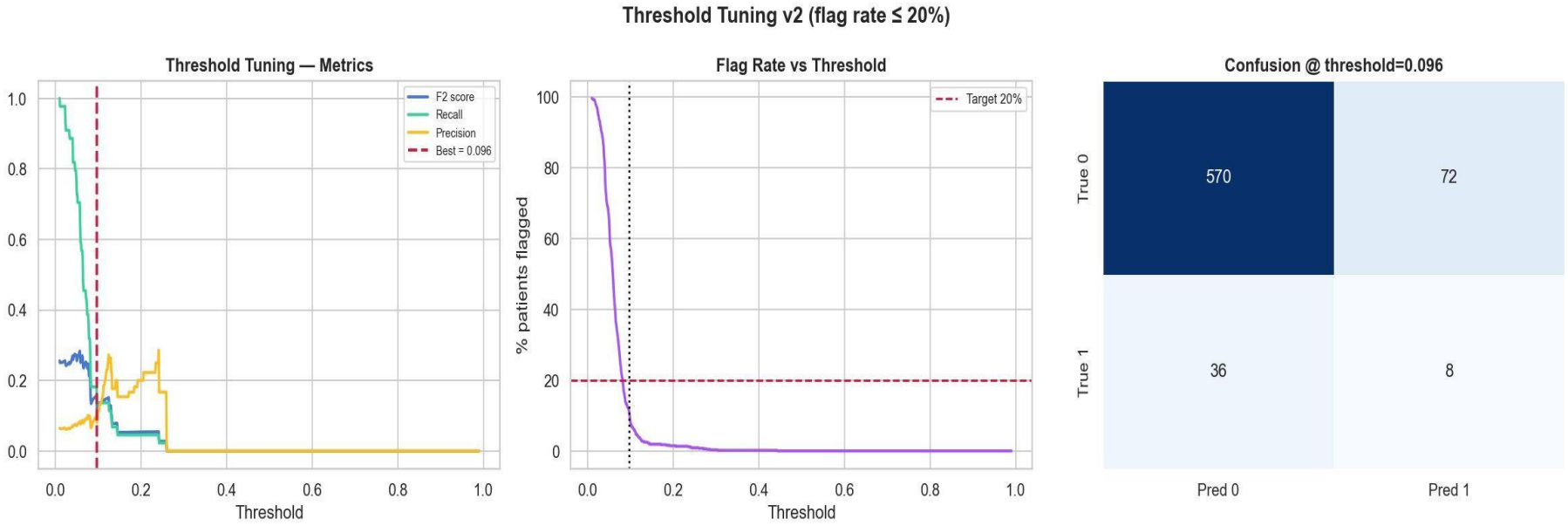
Cluster 2 (AUC 0.40)

Behaviorally distinct cluster struggles — may need cluster-specific calibration.

Equity

Clinical Threshold Tuning

Optimized for ≤20% flag rate — balancing recall against operational burden



Threshold: 0.096

Chosen to keep flag rate ≤20%. Below this, recall is near-perfect but clinic is overwhelmed.

F2-Optimized

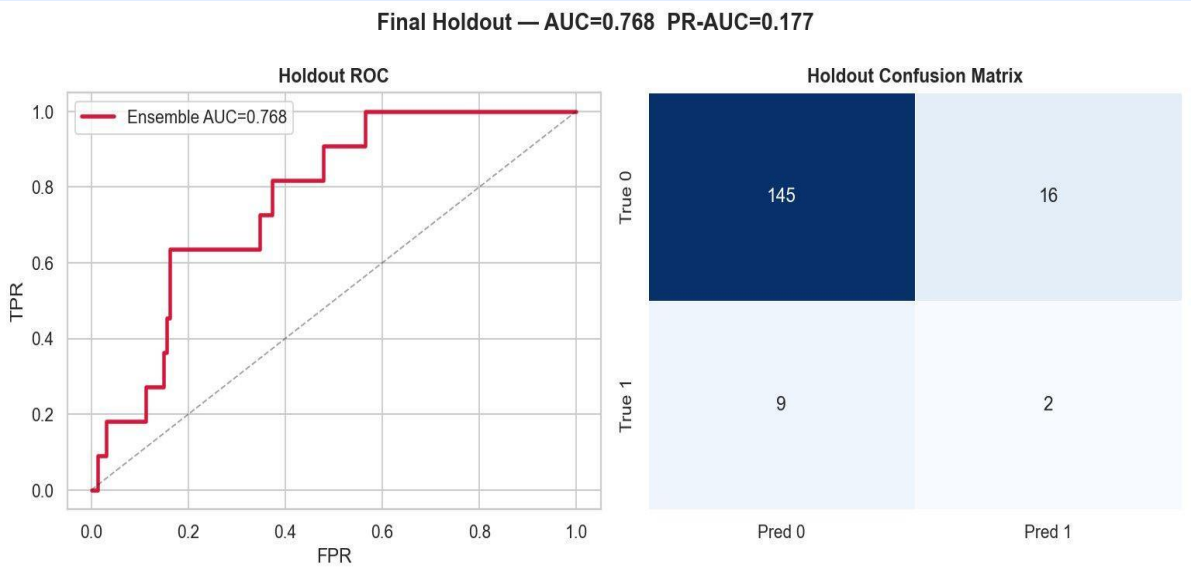
F2 score weights recall 2× over precision — appropriate since missing high-risk patients is costlier than over-referral.

Recall = 73%

At the operating threshold, 8 of 11 holdout positive cases are flagged for follow-up action.

Threshold

Final Holdout Evaluation



Unseen test set (172 patients) — the true measure of real-world performance

Threshold = 0.096 (tuned for ≤20% flag rate, maximizing F2 score to prioritize recall in this high-miss-cost setting)

0.768
Holdout AUC

0.177
Holdout PR-AUC

Confusion Matrix Breakdown

	Pred 0	Pred 1
True 0	145 (TN)	16 (FP)
True 1	9 (FN)	2 (TP)

89%
Specificity

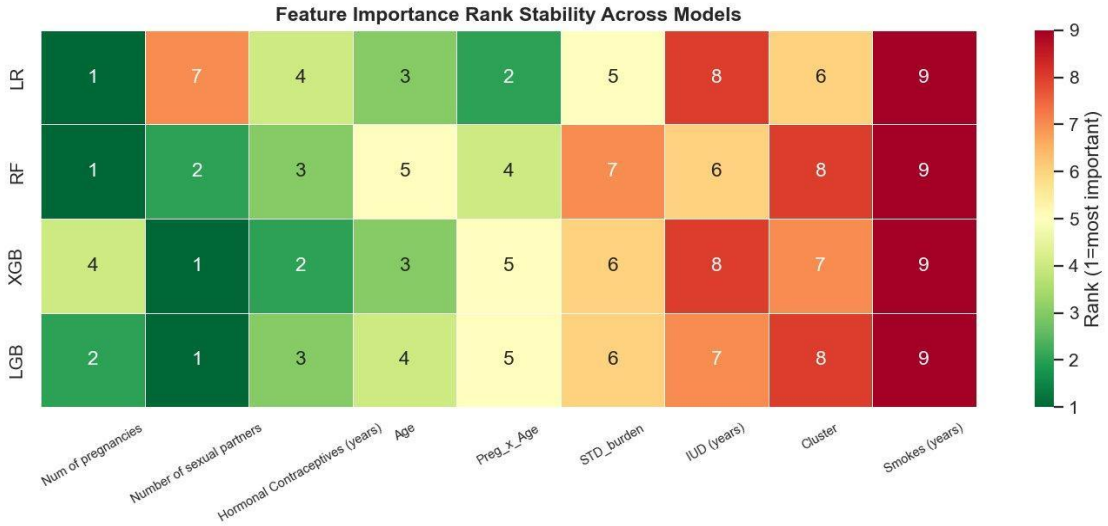
18%
Sensitivity

11%
Precision (PPV)

Holdout Test

Feature Importance: SHAP Analysis

Consistent driver ranking across all four models — builds clinician trust



Feature Rank by Model (1 = most important)				
Num Pregnancies	#1	#1	#4	#2
Sexual Partners	#7	#2	#1	#1
Hormonal Contraceptives	#4	#3	#2	#3
Age	#3	#5	#3	#4
Preg × Age	#2	#4	#5	#5
STD burden	#5	#7	#6	#6
Smoking	#9	#9	#9	#9

Universal drivers

Pregnancies, sexual partners, and hormonal contraceptives rank top-3 in every model — rock-solid signal.

Cross-model stability

The heatmap shows deep green consensus on top features and consistent red for smoking (least important).

Preg × Age interaction

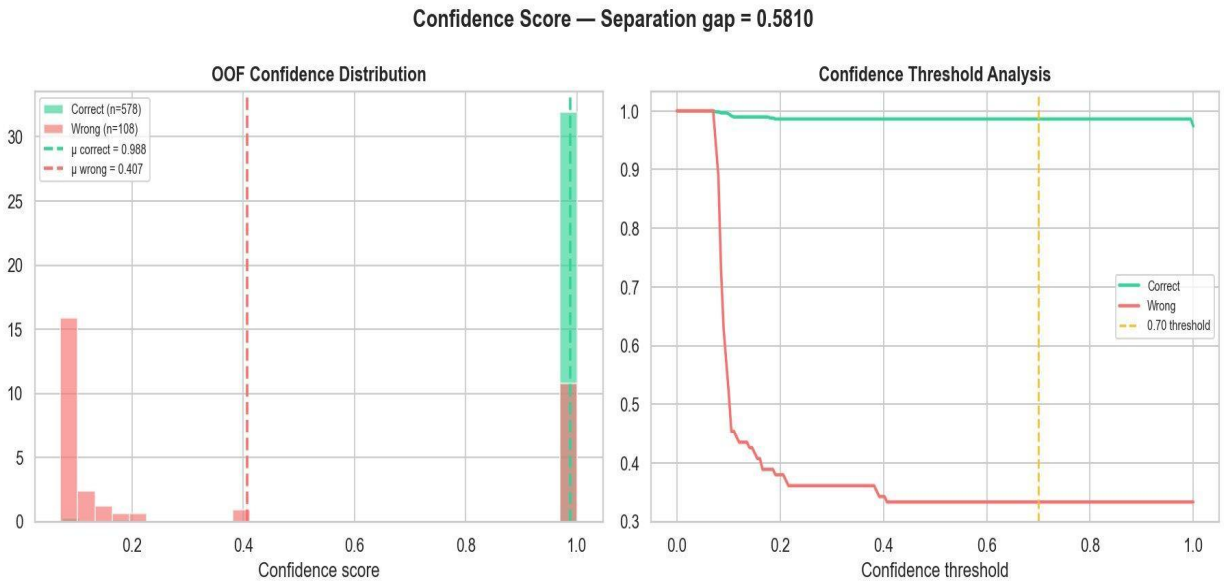
The engineered interaction term captures risk differently across age groups — clinically meaningful.

SHAP = SHapley Additive exPlanations. Values show each feature's marginal contribution to the model output for each patient.

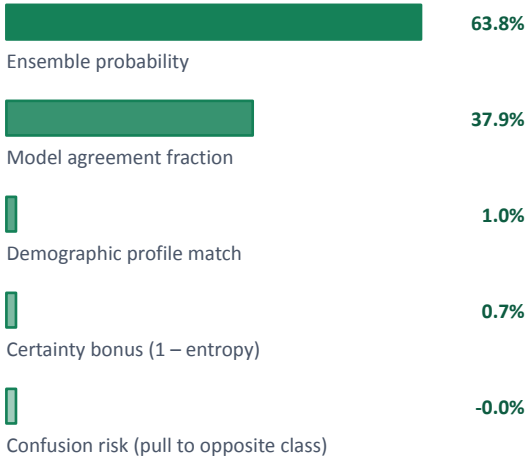
Explainability

Confidence Score System

A composite metric that tells clinicians when to trust the prediction — and when to probe further



Confidence Components



0.581
Separation Gap
(correct vs wrong)

0.988
μ Confidence
(Correct preds)

0.407
μ Confidence
(Wrong preds)

0.70 threshold
At confidence ≥0.70, wrong predictions drop to just ~31% of output — flag low-confidence predictions for review.

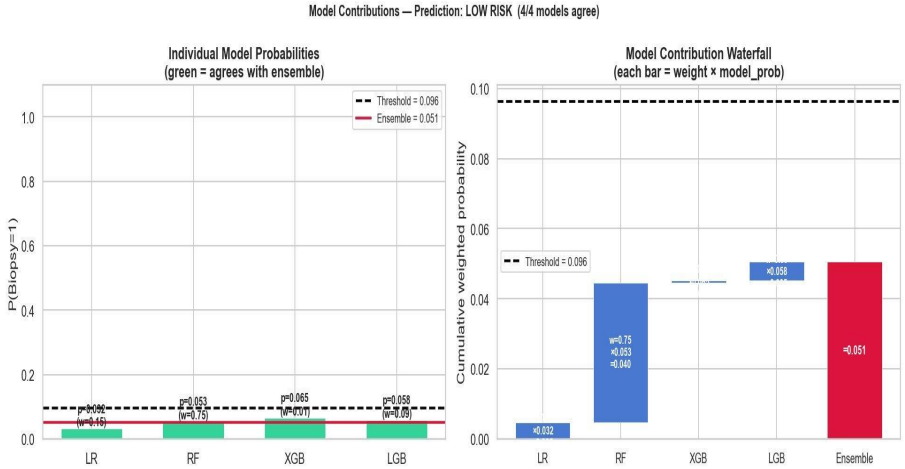
Separation gap of 0.581 means the model "knows what it knows" — correct predictions are highly confident; errors cluster at low confidence.

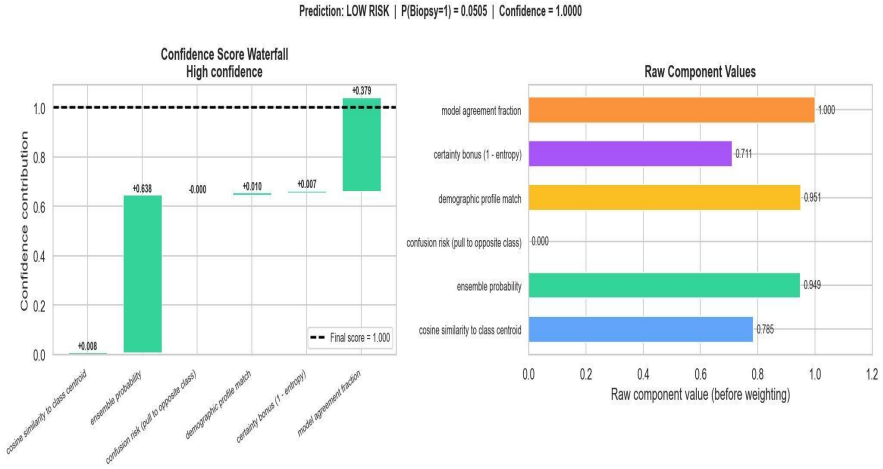
Confidence

Patient Prediction Walkthrough

Example low-risk patient: all four models agree — full transparency to clinician

Prediction: LOW RISK | P(Biopsy=1) = 0.0505 | Confidence = 1.000 | 4/4 models agree





Model Contribution Waterfall (left) · Confidence Score Breakdown (right)

RF dominates (w=0.75)

RF contributes 0.040 weighted prob. LR, XGB, LGB contribute far less — aligns with model weights.

Full model agreement

All 4 models place P(Biopsy) well below the 0.096 threshold — strong consensus → no conflicting signals.

Confidence = 1.0

Model agreement (100%) + ensemble probability (0.949 low-risk facing) + demographic match (0.951) → maximum confidence.

Demo

Patient Position in Population Risk Space

PCA visualization shows where each patient sits relative to high/low/medium risk clusters

3 Risk Clusters

K-means clustering reveals naturally separating patient groups — used as a demographic profile feature in confidence scoring.

PCA for context

PC1 (35.1% var) and PC2 (15.6% var) capture most clinical variation. Clinician sees patient's "neighborhood" at a glance.

True vs. Predicted

Left panel shows ground truth; right shows model score. Near-identical patterns confirm the model learns true risk structure.

Risk Space

Limitations & Path Forward

Honest accounting of current gaps — and what prospective data would unlock

Small holdout positives

Only 11 positive cases in holdout. AUC and sensitivity estimates carry wide confidence intervals — prospective validation needed.

Teen cohort gap

AUC 0.35 for <20 age group (below chance). Features may not capture relevant risk for younger patients.

No temporal features

STI timing and screening history features were >90% missing — dropped. Future data capture could unlock longitudinal risk.

Calibration post-calibration

After isotonic calibration, high predicted probs (>0.6) are empty — model correctly signals low confidence at extremes.

Path Forward

1 Prospective pilot

Deploy as shadow mode (no clinician action required). Collect 500+ prospective cases with confirmed outcomes.

2 Teen-specific model

Separate model or feature set for <20 cohort — consider age-appropriate behavioral predictors.

3 EHR integration

Embed CxRisk in Epic BPA workflow. Auto-populate features from structured chart fields.

4 Live monitoring

Track AUC drift, override rates, and equity metrics monthly. Retrain quarterly with new outcome data.

Roadmap

DSS Structure

Main Components of DSS

DATABASE

- Stores patient demographic data, clinical measurements, standard codes, users, encounter history
- Relational database (PostgreSQL) for quick real-time data retrieval and updates

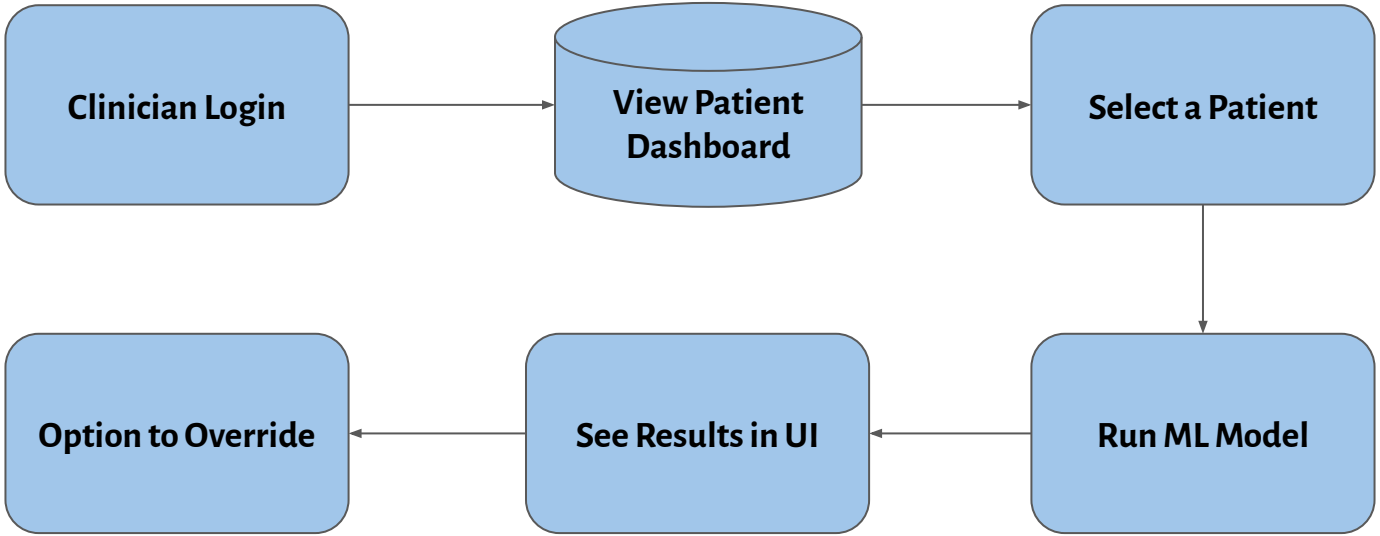
PREDICTIVE ENGINE

- Uses an ML model to predict a patient's individual risk of having cervical cancer, categorizing patients into low, medium, or high risk
- Trained on clinical data, complete with sensitivity and robustness analysis

USER INTERFACE

- Offers interactive interface for clinicians to register new patients, review risk levels and recommendations
- Responsive design (R)
- Displays the risk scores, patient profiles, contributing factors, clinical recommendations, patient summary report
- Features secure login authentication
- Authority lies with clinician in override feature

DSS Flow of Information



Recommendations & Conclusion

Implementation & Deployment Plan (PEIT)

People	Low discretion → required for physicians to use Low sophistication → requires training and transition process
Process	High novelty → new workflow process for clinicians High complexity → integration of predictive model for risk score classification
Group	Span → physicians, administrators Core → directly related to patient satisfaction quality metrics Autonomy → allows for clinician override
Technology	Template → input risk factors directly related to cervical cancer

Pitfalls: Resistance to change, Inertia (slow adoption rate), Non-use (clinicians use their own judgement)

Recommendations for Deployment

- Short-term goals (1-2 years)
  - Encryption, authentication & 2FA
    - Implement end-to-end encryption for all patient data transmissions and enforce two-factor authentication for all DSS users, hardening access to sensitive cervical screening records.
  - Implement user roles
    - Clinicians limited to patient dashboard and risk assessment
    - Administrators limited to population dashboard
    - Introduce a dedicated IT admin role with permissions to add, modify, and deactivate user accounts (reduce administrative burden from clinicians)
  - Automated lab result interpretation
    - Build integration with lab systems to automatically read and parse HPV genotyping results, cytology reports, and biopsy findings, automatically inputting risk factors into DSS model without manual data entry

Recommendations for Deployment

- Long-term goals
  - Patient-facing risk summaries & education
    - Develop a patient-accessible interface presenting simplified risk status, screening history trends, and personalised HPV and cervical health education
  - Upgrade model to address health equity issues
    - Train predictive model on diverse cervical screening datasets to improve model generalizability and to ensure DSS risk predictions perform equitably across different racial, ethnic, and demographic groups

Countermeasures to Pitfalls

Strategies

- Incremental Phasing
  - Begin with a small pilot in a smaller clinic
  - Then scale to a system-wide setting
- Consensus-building styled approach
  - Collaborate with key stakeholders
  - Adjust workflow if it creates bottlenecks in the patient care delivery process

Managing Change

- Rational-empirical approach
  - Clinicians may require evidence that the DSS system provides clinical value
  - Not only a change in the amount of time it takes for diagnosis, but in reducing clinical burden
- Environmental-Adaptive Approach
  - Prevents non-use pitfall
  - Requires clinicians to adapt to change
- Normative-Reeducation Approach
  - Align the goals of the DSS system to the goals of a clinician to their patients

Health & IT Policy Implications

- Data Privacy & HIPAA Compliance:
  - Protecting sensitive health data
  - Cervical screening records carry heightened sensitivity which requires strict compliance with HIPAA and patient consent standards governing the collection, processing, and storage of data
- Cybersecurity Requirements
  - Requires cybersecurity policies in place to protect cervical screening data
  - Methods include end-to-end encryption, access controls, audit logging
- Clinical Standards & Accountability
  - Policies will need to define how DSS outputs are utilized in the clinical workflow
  - Guidelines should consider DSS outputs as recommendations that can be overridden as opposed to strict pathways
- Cervical Cancer Prevention Strategy
  - Strengthening national screening programs
  - Improving early-stage detection through risk-stratified triage, reducing disease progression

Conclusions

- DSS connects clinical context, source data, preprocessing, interpretable risk modeling, workflow design, database architecture, and deployment controls.
- With a clinician-friendly interface, our DSS aims to reduce both mental and alert fatigue faced by physicians.
- To deploy the system on a larger scale, there are many more considerations to take into account such as adding 2FA authentication, role management, and calibrating with reading lab results.
- The DSS supports early risk recognition and consistent follow-up while preserving clinician judgment and patient safety.

Demo

Thank you!

References

"ASCCP Management Guidelines App Quick Start Guide Melissa Gee, MD, DABFM, FAAFP." ASCCP, www.asccp.org/wp-content/uploads/2026/03/ASCCP-App-Quick-Start-Guide-FINAL.pdf. Accessed 24 Apr. 2026.

"Cancer of the Cervix Uteri - Cancer Stat Facts." SEER, 2025, seer.cancer.gov/statfacts/html/cervix.html.

"Cervical Cancer Risk Factors: Risk Factors for Cervical Cancer." American Cancer Society, 1 July 2025, www.cancer.org/cancer/types/cervical-cancer/causes-risks-prevention/risk-factors.html.

"Cervical Cancer." World Health Organization, World Health Organization, 2 Dec. 2025, www.who.int/news-room/fact-sheets/detail/cervical-cancer.

Fernandes, Kelwin, et al. "Cervical Cancer (Risk Factors)." UCI Machine Learning Repository, 2 Mar. 2017, doi.org/10.24432/C5Z310.

"Guidelines: ASCCP Risk-Based Management Consensus Guidelines for Abnormal Cervical Cancer Screening Tests and Cancer Precursors." ASCCP Risk-Based Management Consensus Guidelines for Abnormal Cervical Cancer Screening Tests and Cancer Precursors | BD Women's Health, womens-health-solutions.bd.com/resources/articles-publications/guidelines-asccp-risk-based-management-consensus-guidelines. Accessed 24 Apr. 2026.

Perkins, Rebecca B, et al. "2019 ASCCP Risk-Based Management Consensus Guidelines for Abnormal Cervical Cancer Screening Tests and Cancer Precursors." Journal of Lower Genital Tract Disease, U.S. National Library of Medicine, 2 Apr. 2020, pmc.ncbi.nlm.nih.gov/articles/PMC7147428/.

"User Guide: ASCCP Management Guidelines App." American Society for Colposcopy and Cervical Pathology (ASCCP), 13 Mar. 2026, www.asccp.org/mobile-app/user-guide/.

Zhang, Shaokai, et al. "Cervical Cancer: Epidemiology, Risk Factors and Screening." Chinese Journal of Cancer Research = Chung-Kuo Yen Cheng Yen Chiu, U.S. National Library of Medicine, 31 Dec. 2020, pmc.ncbi.nlm.nih.gov/articles/PMC7797226/.